1. VRC(Vertical Redundancy Check)

Even parity: No of ones should be even. If not, make them even by making redundant bit 1.

Send: 111001 0__

Receiver: 1110010→ even parity

1110 ***10*** 0 Received → even parity

Error is not detected even there is an error of two bits.

000001 0_: Received

Odd parity, so error is detected.

So, VRC can detect all odd bits error.

2. LRC:

11100111 11011101 00111001 10101001

Even parity

11100111 _

11011101 _

00111001 _

10101001 __

10101010 _

Send: 11100111 11011101 00111001 10101001 10101010

Received: 11100111 11011101 00111001 10101001 10101010

11100111

11011101

00111001

10101001

10101010

It can detect burst error of n bits (n is length of one section).
Burst error of more than n bits is also detected with a very high probability. For eg two bit changes in same column cannot be detected.

3. CRC (Cyclic Redundancy Check) : Modulo 2 division (XOR Gate)

1101 |100100

$1*X^3+ 1*X^2+ 0*X^1+ 1*X^0$:1101

Q. 110111101

```
          111101
110   100100 000
1
      1101
      *1000
       1101
       *1010
        1101
        *1110
         1101
         *0110
          0000
          *1100
           1101
           001
```

Send: 100100 001

Receiving End:

```
1101 )100100 001(  111101
      1101
      *1000
       1101
       *1010
        1101
        *1110
         1101
         *0110
          0000
          *1101
           1101
           000
```

As remainder is 0, so there is no error.

Polynomial should be selected to have at least the following properties:

1. **It should be not be divisible by x.**
2. **It should be divisible by (x+1)**

Performance:

1. **CRC can detect all burst error that affect an odd number of bits.**
2. **CRC can detect all burst error of length less than or equal to the degree of the polynomial.**
3. **CRC can detect with a very high probability burst error of length greater than the degree of the polynomial.**

4. Checksum:

10101001 00111001

1 1 1 1

10101001

+00111001

11100010 (addition using 1's Complement)

Complement the result to get the checksum

Checksum: **00011101**

Transmitted Data: 10101001 00111001 00011101

1 11 1

10101001

00111001

00011101

11111111

Complement the result

00000000

If all bits are 0 that means there is no error.

Q: Error burst of 5 bits has occurred.

10101*111 11*111001 00011101

10101111

11111001

00011101

11000101

+1

11000110

Complement

00111001

Hence there is an error.

Error Correction

Hamming code is use for error correction.

Single bit error correction.

For 7 bits data, we have 8 possibility:

1. **No error (000)**
2. **Error at position 1 (001)**
3. **Error at position 2 (010)**
4. **Error at position 3**
5. **…….4**
6. **…….5**
7. **…….6**
8. **…….7 (111)**

$2^3=8 \rightarrow$ 3 bits

To represent this we need 3 bits (Redundant bits)

But, the error can be at redundant bits also, so more no of bits are req. Which can be calculated as follows:

No of redundant bits (r): Where, m are the no of bits in data

$2^r >= m+r+1$

For eg. M=7,

Put r=3

$2^3 >= 7+3+1$

8>=11 Incorrect

Put r=4

$2^4 >= 7+4+1$

16>=12

So 4 will be the redundant bits, for m=7

M=11

Put r=4

2^4 >=11+4+1

16>=16

M=12

Put r=4

2^4 >=12+4+1

16>=17 (Incorrect)

M=12

Put r=5

2^5 >=12+5+1

32>=18

R=r1, r2, r3, r4

	r4	r3	r2	r1
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10	1	0	1	0
11	1	0	1	1
12	1	1	0	0

13	1	1	0	1
14	1	1	1	0
15	1	1	1	1

r1→1,3,5,7,9,11,13,15

r2→ 2,3,6,7,10,11,14,15

r3→ 4,5,6,7,12,13,14,15

r4→8,9,10,11,12,13,14,15

These are the positions where we will check the even/odd parity for each redundant bit.

Put redundant bits at positions 2^n where n=0,1,2,4….

D1	D2	D3	r4	D4	D5	D6	r3	D7	r2	r1
11	10	9	8	7	6	5	4	3	2	1

D=1001101

1	0	0		1	1	0		1		
11	10	9	8	7	6	5	4	3	2	1

Even Parity:

r1 →3 1's→1 → 4 1's So even parity

r2→4 1's→ 0

r3→ 2 1's →0

r4→ 1

1	0	0	1	1	1	0	0	1	0	1
11	10	9	8	7	6	5	4	3	2	1

Data to be transmitted: 10011100101

1	0	0	1	**0**	1	0	0	1	0	1
11	10	9	8	**7**	6	5	4	3	2	1

Data received: 10010100101

r1→1,3,5,7,9,11,13,15

r2→ 2,3,6,7,10,11,14,15

r3→ 4,5,6,7,12,13,14,15

r4→8,9,10,11,12,13,14,15

r1→1

r2→ 1

r3→1

r4→0



0111→Error is at 7th position

1	0	0	**0**	1	1	0	0	1	0	1
11	10	9	**8**	7	6	5	4	3	2	1

Data to be transmitted: 10011100101

Data Received : 10001100101

r1→1,3,5,7,9,11,13,15

r2→ 2,3,6,7,10,11,14,15

r3→ 4,5,6,7,12,13,14,15

r4→8,9,10,11,12,13,14,15

r1→ 0

r2→ 0

r3→ 0

r4→ 1

1000 → Error is at 8 position

Data Link Layer Design issues:

1 Framing

2 Physical Addressing

3 Error Control

4 Flow Control

5 Access Control

Various services provide by the Data Link layer to Network Layer

1. **Unacknowledged connectionless service**
2. **Acknowledged connectionless service**
3. **Acknowledged connection-oriented service**